

LAGHOUAT, Algeria

WMO: 605450

Lat: 33.764N

Lon: 2.927E

Elev: 2510

StdP: 13.41

Time Zone: 1.00 (E01)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.2	33.1	18.0	14.9	53.8	21.1	17.3	52.2	37.9	52.6	32.3	50.1	1.0	240	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.3	106.1	68.3	104.2	67.8	102.1	67.4	72.9	96.9	71.1	95.8	69.7	94.4	13.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	105.4	81.7	64.1	98.3	79.9	62.3	92.0	78.6	38.1	96.8	36.4	95.6	35.2	94.3	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.8	25.7	22.5	25.8	109.4	2.5	1.3	24.0	110.3	22.6	111.0	21.2	111.8	19.4	112.7		
(5)			24.3	76.0	2.2	2.8	22.7	78.0	21.4	79.6	20.2	81.2	18.6	83.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.8	47.3	49.8	56.7	64.1	72.8	82.4	89.8	87.2	78.0	68.0	55.5	48.5		
	DBStd	15.73	4.62	5.85	6.00	6.69	6.59	5.97	3.89	4.35	5.96	5.83	5.37	4.84		
	HDD50	284	109	67	13	0	0	0	0	0	0	0	10	85		
	HDD65	2183	549	427	262	96	12	0	0	0	1	35	291	510		
	CDD50	6408	24	61	222	424	707	973	1233	1154	839	557	174	40		
	CDD65	2833	0	1	6	70	253	524	768	689	390	127	5	0		
	CDH74	40020	0	15	160	899	3097	7402	12327	10252	4512	1293	61	2		
	CDH80	23624	0	2	22	276	1447	4406	8259	6472	2303	430	7	0		
(14) Wind	WSAvg	8.8	8.1	10.3	10.5	11.2	10.7	10.1	8.8	7.9	7.7	6.3	7.5	6.9		
(15) Precipitation	PrecAvg	6.6	0.5	0.4	0.5	0.6	0.6	0.4	0.2	0.5	0.9	0.7	0.6	0.6		
	PrecMax	11.3	2.5	1.7	2.0	3.4	2.1	1.4	1.2	3.1	3.1	3.2	3.7	2.8		
	PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.4	0.6	0.4	0.6	0.7	0.6	0.4	0.3	0.6	0.7	0.8	0.9	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.2	75.5	82.8	91.1	98.7	105.5	109.5	107.7	101.3	93.4	80.1	71.2		
		MCWB	51.7	53.1	56.4	59.6	63.7	68.2	67.6	68.3	65.7	65.0	58.0	54.0		
	2%	DB	65.8	70.0	78.6	86.4	94.8	102.6	107.5	105.4	97.2	88.7	74.9	66.3		
		MCWB	50.2	51.7	54.7	58.5	63.3	66.5	68.2	67.8	66.3	62.5	56.5	52.5		
	5%	DB	62.4	66.1	74.9	82.8	91.4	100.2	105.6	102.6	94.9	84.6	70.8	62.6		
		MCWB	49.0	50.2	53.7	57.5	62.6	66.3	68.5	67.4	66.0	62.0	54.9	50.6		
	10%	DB	59.0	62.3	71.3	79.1	88.0	97.0	103.6	100.4	91.6	80.9	67.6	59.0		
		MCWB	47.7	48.5	52.1	56.7	61.2	65.9	68.1	67.4	65.5	61.1	53.9	49.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	56.1	59.2	63.8	69.4	75.3	76.4	74.7	72.0	68.1	61.9	55.8		
		MCDB	65.9	71.9	76.0	83.2	87.8	96.4	100.4	97.8	92.2	82.3	74.8	65.2		
	2%	WB	51.7	53.2	56.7	60.9	66.4	71.6	73.0	72.1	69.8	65.6	59.0	53.3		
		MCDB	62.8	66.6	74.3	80.8	87.4	94.6	98.5	96.4	88.8	79.9	69.8	63.2		
	5%	WB	49.7	51.1	55.0	59.0	64.6	68.8	71.2	70.4	68.4	64.1	56.6	51.5		
		MCDB	60.6	63.4	71.6	77.0	85.7	93.2	98.3	94.4	86.7	78.9	67.4	61.0		
	10%	WB	48.0	49.1	53.1	57.5	62.7	67.1	69.7	69.2	67.2	62.7	54.6	49.6		
		MCDB	58.0	60.6	68.4	75.9	83.4	92.1	97.7	93.0	85.3	76.6	65.2	57.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.1	21.7	23.5	23.9	24.6	25.4	26.3	25.4	23.5	23.1	21.4	20.8		
		MCDBR	28.6	28.5	30.0	29.1	28.4	27.9	28.2	27.6	27.1	27.5	26.7	27.0		
		MCWBR	16.4	14.3	13.6	11.9	10.8	10.1	9.4	9.5	9.1	10.6	12.7	15.0		
		MCDBR	26.2	26.3	26.9	25.8	25.8	25.7	26.2	25.5	24.3	23.1	23.0	23.4		
	5% WB	MCWBR	15.8	14.1	12.9	11.7	11.1	11.6	10.7	9.7	9.7	10.4	12.1	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.311	0.389	0.426	0.478	0.510	0.570	0.533	0.487	0.417	0.335	0.294		
	taud		2.444	2.357	2.105	2.023	1.901	1.829	1.706	1.793	1.906	2.077	2.325	2.422		
	Ebn at Noon		298	298	279	273	260	251	235	241	246	254	272	285		
	Edh at Noon		28	34	48	55	62	67	75	68	58	45	31	27		
(44) All-Sky Solar Radiation	RadAvg		1063	1362	1765	2106	2327	2505	2448	2204	1801	1461	1119	961		
	RadStd		67	81	85	93	104	72	62	72	84	98	55	62		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.56	N/A	-2.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page